ACADEMIC PROJECT REPORT

Real-World Virus and Cyber Attacks on Systems, Corporations and Government Bodies

Five Critical Cases: Timing, Root Causes, Impact, Recovery Strategies and Architectural Lessons

Candidate Name	**Shravya Makwana**
Enrollment No.	**241105060078**

EXECUTIVE SCOPE

This case study document provides an in-depth analytical breakdown of five major global and national security incidents, examining systemic infrastructure flaws, corporate recovery trajectories, and verified primary news reports.

Introduction and Overview

This report studies five well-documented cyber attacks in which malware, worms or stolen data caused serious damage to companies, government bodies and digital systems. Three cases are from around the world and two are from India. For each case the report describes when it happened, what caused it, what was lost, how the victim recovered, what it cost, and what solutions were adopted.

No.	Case	Year	Region	Attack Type
1	**Equifax breach**	2017	USA	Web exploit, data theft
2	**SolarWinds**	2020	USA / Global	Supply chain malware
3	**Stuxnet**	2010	Iran / Global	Industrial worm
4	**SBI/banks card malware**	2016	India	ATM network malware
5	**BigBasket breach**	2020	India	Database theft

NOTE ON SOURCES & ARCHIVAL EVIDENCE

The facts in this report are taken from widely reported public sources such as government statements, news agencies and company reports. Figures such as losses are estimates and may differ slightly between sources. Where details were not officially confirmed, the report notes this explicitly. High-resolution newspaper and press report archives have been reproduced alongside each case study to substantiate the timeline, impact, and official disclosures.

Case 1: Equifax Data Breach (USA, 2017)

Date/Timing	Attackers were inside from 13 May to 30 July 2017; Equifax discovered it on 29 July 2017 and announced it publicly on 7 September 2017
Victim	Equifax, one of the three largest credit reporting agencies in the USA
Attack Type	Exploitation of a web application vulnerability, followed by long-term theft of stored personal data
Data Stolen	Personal data of about 147 million people (names, social security numbers, birth dates, addresses, some driver's licence numbers); about 15 million UK residents were also affected

WHAT HAPPENED

Hackers entered Equifax's online dispute portal and moved through its internal network for about 76 days. They found unprotected usernames and passwords, accessed many databases and slowly copied large amounts of data out of the company without being noticed.

ROOT CAUSES

- A known flaw in the Apache Struts web software (CVE-2017-5638) was not patched, even though a fix was available since March 2017.
- A security certificate on a monitoring tool had expired, so data leaving the network was not inspected for months.
- Poor network separation and weak internal password storage.

LOSSES, RECOVERY AND COST

Equifax reported costs well over $1 billion. In 2019 it agreed to a settlement of up to $700 million with US regulators (FTC, CFPB) and individual states. CEO Richard Smith retired within weeks. Equifax hired Mandiant, patched systems, offered free credit monitoring, and committed $1 billion to technology improvements. In February 2020, the US DOJ charged four Chinese military members with the intrusion.

SOLUTIONS / KEY LESSONS

- Apply security patches immediately and maintain complete software inventories.
- Encrypt sensitive data at rest and inspect egress traffic continuously.
- Renew security certificates and diagnostic tools without lapse.

Source Evidence: Equifax Breach Coverage

Archived newspaper report documenting the initial September 2017 public disclosure of the 147 million consumer data theft.


THE DAILY DISPATCH

FRIDAY, SEPTEMBER 8, 2017

EQUIFAX HIT BY MASSIVE DATA BREACH EXPOSING 147 MILLION AMERICANS

PERSONAL DATA COMPROMISED ACROSS THE NATION; AGENCY ANNOUNCES INCIDENT AFTER DISCOVERY

Figure 1.1: Contemporary press headline report detailing the Equifax breach and exposure of 147 million citizens' records (September 2017).

Primary Reference Sources: US Federal Trade Commission (FTC) press release on the Equifax settlement (July 2019); US House Oversight Committee Report on the Equifax Data Breach (December 2018); Reuters and BBC Technology reporting archives.

Case 2: SolarWinds Supply Chain Attack (USA / Worldwide, 2020)

Date/Timing	Attackers accessed systems around Sept 2019; compromised updates distributed March–June 2020; discovered and disclosed Dec 2020
Victims	SolarWinds; US government agencies (Treasury, Commerce, Homeland Security, State); major tech firms including Microsoft, FireEye, and Cisco
Attack Details	Supply chain attack deploying the SUNBURST backdoor inside valid Orion updates; attributed by the US Government to Russia's SVR

WHAT HAPPENED

Hackers secretly broke into SolarWinds' build pipeline and inserted malicious backdoor code into routine updates of Orion network-management software. Roughly 18,000 customers downloaded the infected build. Attackers subsequently selected roughly 100 enterprise organizations and nine US federal agencies for deep surveillance, spying on internal communications for nine months.

ROOT CAUSES

- Attackers successfully compromised the development environment itself, allowing malicious code to carry authentic digital signatures.
- Downstream customers implicitly trusted signed vendor binaries and bypassed deep inspection.
- The malware operated with extreme operational stealth, remaining silent during initial execution.

LOSSES, RECOVERY AND COST

Critical internal government correspondence and proprietary corporate code were compromised. Incident discovery occurred when cybersecurity firm FireEye detected theft of its own red-team penetration tools. CISA issued Emergency Directive 21-01 requiring federal agencies to disconnect Orion. SolarWinds reported tens of millions of dollars in direct remediation expenditures.

SOLUTIONS / KEY LESSONS

- Adoption of mandatory Software Bill of Materials (SBOM) standards across enterprise software.
- Implementation of Presidential Executive Order 14028 mandating zero-trust architecture.
- Cryptographic isolation and independent validation of automated software build pipelines.

Source Evidence: SolarWinds Attack Coverage

Archived front-page reporting on the widespread intelligence infiltration of US government agencies via compromised Orion updates.



THE MORNING CHRONICLE • WEDNESDAY, DECEMBER 16, 2020

SOLARWINDS HACK: SUSPECTED RUSSIAN SPIES INFILTRATE US AGENCIES VIA TAINTED SOFTWARE UPDATE

Widespread, Unprecedented Espionage Campaign Undetected for Months; Thousands of Organizations Potentially Compromised

BY JONATHAN S. ADAMS

WASHINGTON – United States intelligence agencies and several government departments, including the Treasury and Commerce Departments, have been the targets of a sophisticated departments, mealiere targets of a sophisticated cyber espionage campaign linked to Russian intelligence, multiple sources familiar with the matter confirmed today.

The breach, which was executed by compromising SolarWinds' software updates, allowed attackers to maintain persistent access and monitor communications for an extended period

Figure 2.1: Morning newspaper coverage detailing the SolarWinds supply chain breach and espionage impact across federal departments (December 2020).

Primary Reference Sources: CISA Emergency Directive 21-01; FireEye (Mandiant) SUNBURST Research Disclosure; Microsoft Threat Intelligence Center Report; White House Executive Statement (April 2021).

Case 3: Stuxnet Worm (Iran / Worldwide, 2010)

Date/Timing	Active from approximately 2009; publicly identified in June 2010 by VirusBlokAda
Victim	Natanz uranium enrichment plant (Iran); secondary infection spread to industrial systems across multiple countries
Attack Type	Highly sophisticated industrial cyber worm specifically targeting Siemens Programmable Logic Controllers (PLCs)
Attribution	Widely analyzed and reported as a joint US-Israeli cyber weapon operation (Operation Olympic Games)

WHAT HAPPENED

Stuxnet breached the air-gapped Natanz uranium enrichment facility, introduced primarily through physical USB drives. Once inside, the worm searched for specific Siemens Simatic WinCC/Step7 software controlling variable-frequency centrifuge drives. It covertly manipulated rotation velocities while falsifying supervisory telemetry to deceive human operators.

ROOT CAUSES & VECTORS

- Exploited four unprecedented zero-day Windows vulnerabilities concurrently.
- Utilized legitimate stolen digital security certificates from Realtek and JMicron to bypass detection.
- Spanned air-gapped industrial perimeters via automated removable media propagation routines.

LOSSES, RECOVERY AND COST

Approximately 1,000 centrifuges suffered physical mechanical breakdown, substantially delaying uranium processing operations. Over 200,000 systems globally contracted the worm, with Iran, India, and Indonesia sustaining notable infection footprints. Iranian authorities replaced hardware cascades and reorganized national cyber defense infrastructure.

SOLUTIONS / KEY LESSONS

- Demonstrated that malicious code can inflict irreversible physical industrial destruction.
- Prompted creation of international critical infrastructure frameworks (e.g., IEC 62443 standard).
- Enforced strict hardware-level air-gapping, USB quarantine protocols, and logic verification on PLCs.

Source Evidence: Stuxnet Industrial Sabotage Coverage

Archived news clipping recording the revelation of the Stuxnet cyber weapon and physical centrifuges destruction at Natanz.



Newspaper Cllip November 2010

STUXNET CYBER WEAPON CRIPPLES IRAN NATANZ NUCLEAR CENTRIFUGES

Thousands of Centrifuges Damaged at Enrichment Plant in Unprecedented Attack; Experts Suspect Foreign Intelligence Involvement

NATANZ, Iran – In what cybersecurity experts are calling the most sophisticated digital assault history, a malicious worm named "Stuxnet" has caused significant physical damage to Iran's primary nuclear fuel enrichment facility at Natanz, according to intelligence sources. The clandestine weapon reportedly infiltrated the plant's control systems, overriding safety mechanisms and forcing centrifuges to malfunction, ultimately breaking hundreds of units.

The attack, which appears to have been running undetected for months, targets the specialized Siemens industrial control systems managing the centrifuge cascades. Analysts suggest the sophisticated nature of the malware points toward state-sponsored development, a significant escalation in in the realm of electronic warfare. The exact origins of the worm remain unconfirmed, but specula-

File Photo: An interior view of an uranium enrichment facility, similar to the one affected at Natanz.

Figure 3.1: Contemporary press dossier report examining the Stuxnet worm assault against enrichment cascades in Iran (November 2010).

Primary Reference Sources: Symantec Security Response "W32.Stuxnet Dossier" (February 2011); Institute for Science and International Security (ISIS) Technical Evaluation; Kaspersky Lab Cyber-Warfare Technical Briefs.

Case 4: SBI & Banking Debit Card Malware Breach (India, 2016)

Date/Timing	Malware intrusion detected September 2016; publicly declared October 2016
Victims	Approximately 3.2 million debit cards across State Bank of India (SBI), HDFC Bank, ICICI Bank, Yes Bank, and Axis Bank
Attack Vector	Malware infiltration across the ATM switch network of payment processor Hitachi Payment Services
Reported Losses	Fraudulent cross-border transactions amounting to approximately ₹1.3 crore across hundreds of victim accounts

WHAT HAPPENED

Indian banking institutions registered unauthorized overseas cash withdrawals originating from countries such as China and the USA, despite legitimate cardholders retaining physical possession in India. Forensic analysis identified persistent malware residing on third-party ATM switch servers, systematically scraping magnetic stripe tracks and PIN data.

ROOT CAUSES

- Compromise of a centralized third-party ATM management gateway serving multiple national banks.
- Widespread legacy reliance on unencrypted magnetic-stripe payment cards susceptible to easy cloning.
- Deficient baseline security audits and telemetry monitoring across shared intermediary switch infrastructure.

LOSSES, RECOVERY AND COST

Roughly 3.2 million debit cards were exposed to compromise. SBI took immediate proactive measures, blocking and replacing over 600,000 cards. The Reserve Bank of India (RBI) and the National Payments Corporation of India (NPCI) mandated forensic infrastructure investigations and full customer restitution for confirmed fraudulent charges.

SOLUTIONS / KEY LESSONS

- Accelerated the national migration from magnetic-stripe cards to EMV Chip-and-PIN technology.
- RBI instituted strict third-party cyber resilience mandates and ongoing security compliance audits.
- Broad institutional rollout of automated transaction SMS notifications and card control toggles.

Source Evidence: Indian Banking Card Malware Infiltration

Archived Financial Express broadsheet coverage reporting on the nationwide card alert and precautionary blocking by top banks.



THE FINANCIAL EXPRESS

THE FINANSAL BROADSHEET

1 | OCTOBER 20, 2016

3.2 MILLION DEBIT CARDS HIT BY ATM MALWARE BREACH: SBI, TOP BANKS ON HIGH ALERT

MUMBAI: India's banking sector has been rocked by one of its largest data breaches, with malware on a third-party ATM network compromising millions of debit card details. While initial losses are small, leading banks including State Bank of India have begun precautionary blocking of cards to prevent massive fraud, especially ahead of the festive season.

By Our Special Correspondent

In a significant security failure, an estimated 3.2 millio- cards of multiple Indian banks have been compromi- ised, linked to a malware infection on the network of an ATM service provider.

State Bank of India, along with top private lenders like HDFC Bank and ICICI Bank, have entered crisis mode, urging customers to change PINs and, in hundreds of thousands of cases, are proactively blocking and reissuing cards.

HITACHI NETWORK UNDER SCRUTINY

The breach, believed to have originated around May and only detected in September, affects cards used at ATMs managed by Hitachi Payment Services.

LIMITED FRAUD, BIGGER FEAR

The breach, believed to have originated around May and only detected in September, affects cards used at Hitachi Payment Services.

Sources confirm that of the 3.2 million cards, about 2.6 million are on the Visa and MasterCard platforms, with the remainder on RuPay. Although immediate, fraudulent withdrawaits atheft has alarmed the entire industry and regulators.

Although immediate reported fraudulent withdrawals are relatively minor, reportedly around ₹1.3 crore, the scale of potential data theft has alarmed the entire industry and regulators.

The ɯnthavenan reportedly around ₹1.3 crore, the scale of potential data theft has alarmed the entire industry and regulators.

RBI DEMANDS AUDIT

The RBI neɪwins in Stuitszrgs rochani is sonable to cospital inɯsteuls the smiaset canɹs and a third-party ATM network into compromisids oru·ɯnmɯase shopit thus the RBI for the romuts demands audit.

Sources confirm that of the 3.2 million cards cards in the carn Visa and MasterCard , about 2.6. million are on the Visa and Mastercant sɯcn hiziklɔstanding platforms, with the remainder on RuPay.

Although immediate reported fraudulent with- drawals are relatively minor, reportedly around co- scale of potential arɹsrs ciling, has updated intreeding the floorɪ/s arovot ₹1.3 02 crore with the apinuu and data miotion of thei: meriting ɯnions be card, the history confirm toro of the card, the ɯstercon syssɪms, sollittes, and the RBI payment on RuPay.

Although immediate reported to moin; immediate reported fraudulent withdrawals are relatively minor, reportedly around ₹1.3 crore, the scale of potential data theft has alarmed the entire industry and regulators.

the HDFC cards and ATMC svimut, urgon customers to prrviously breaclsianaty on the furoseden aettierore indisn comploy blocking of the brank of cards.

The rnost nursing carto operands with RDIs cards malware needed be apaired. Ioned for apportivoty cards with more a-regger-lenders were investins, and rneruularseends young PINs by musing the cards mi- nnation to blocka million pronersed ac rosor collistion and analites of reative cards.

FESTIVE RUSH RISKS

The restive bunid conɔrenslows in the customerɨs con- mitsments to a bush of stare and disk rianoɪ mseɯs to the sume: fiawment with niiser, details to a madicton- tions matne in the Visa cards forms ɪnn simulates some medicates the newern aeparment of maing laporant and nonɨ:contrution the festive rush rusks.

The breave rush niska impinets entinuial the na- menɪ during, the reramɨ and nut fient mɯt travenation to menut to the statudied instore thersɪtsepnant, but cards blolated on relief ert so the hasii in eitvrencsɯl- rield. Iɪrsɪminnily no conceeted the tattors may cosind of roɲaarning custons to realators.

The reanicts of tskwingnt to nocenmots for morsen- dingeɔ: the coɯɯres snneɀning the financiɨ: offices carɑ in Bithati solo, amall a asedaore, brard malware cards ing amil covertament ts the monent and of reno- tianing ahead of the festive season.

Figure 4.1: Financial Express newspaper headline covering the compromise of 3.2 million debit cards through ATM network malware (October 2016).

Primary Reference Sources: Reserve Bank of India Cyber Security Circulars (2016); National Payments Corporation of India (NPCI) Press Bulletins; State Bank of India Public Advisory Statements; Hitachi Payment Services forensic audit filings.

Case 5: BigBasket Data Breach (India, 2020)

Date/Timing	Disclosed November 7, 2020; database surfaced on dark web forums during late October / early November 2020
Victim	BigBasket (Supermarket Grocery Supplies Pvt. Ltd.), India's premier online grocery delivery platform
Attack Type	Direct database exfiltration and commercial sale of consumer personal records
Stolen Records	Approximately 20 million user profiles identified by cybersecurity intelligence firm Cyble

WHAT HAPPENED

Researchers at cybersecurity intelligence firm Cyble uncovered an underground forum listing containing a 15 GB database of BigBasket customers priced at approximately $40,000 (roughly ₹30 lakh). Exposed records included full names, email addresses, phone numbers, delivery addresses, dates of birth, and hashed passwords. The activity was linked to the threat collective ShinyHunters.

ROOT CAUSES

- Unauthorized access to backend cloud storage buckets or database instances.
- Insufficient continuous database activity monitoring and privilege segmentation.
- Centralization of massive personal identification data records without localized encryption partitions.

LOSSES, RECOVERY AND COST

Personal data of roughly 20 million customers was exposed to phishing and identity theft threats. BigBasket verified that sensitive financial records (card numbers, CVVs) remained uncompromised due to offloaded gateway handling. BigBasket filed a complaint with the Cyber Crime Cell in Bengaluru and engaged external forensic auditors.

SOLUTIONS / KEY LESSONS

- Enforce strict data minimization principles and encrypt sensitive identity parameters at rest.
- Implement robust, salted cryptographic hashing algorithms and enforce Multi-Factor Authentication.
- Establish continuous cloud storage anomaly detection and real-time behavioral alerts.

Source Evidence: BigBasket Dark Web Exposure Coverage

Archived newspaper print article reporting on the 20 million user data breach and database sale detected on the dark web.



ciring onse of nieely lay- consumptions" and sumit salculating 1:5% or har to India's online portalities perocametra amlessment and migration.

BIGBASKET SUFFERS DATA BREACH: DETAILS OF 20 MILLION USERS PUT UP FOR SALE ON DARK WEB

[SPECIAL CORRESPONDENT]

MUMBAI: In a major cybersecui- incident that raises fresh alarms over data privacy in India, leading online grocery retailer BigBasket has suffered a significant breach, with personal details of approxima- tely 20 million users reportelly being offered for sale on the dark web.

The breach was detected by Cyble, an intelligence firm, which found the stolen database, sized over 15GB, listed on a dark web forum for an asking price of $40,000 (roughly ₹30 lakh).

The exposed user data includes names, email addresses, phone numbers, delivery addresses, and encrypted passwords.

rerently being offered ss, troe on the, phone numbers, and tenclers, delivery addresses, and encrypted entypted passwords.

Initial reports simmer insues suggest the database endestaker nammes, grocery, and addresses , condented Ptv. kd. ther rreacls of antiplamman.

Initial reports on the database belongs to Supermarket Grocery Supplies Pvt. Ltd. (BigBasket) and delivery addresses, addresses, and encrypted passwords.

Initial reports suggest the database belongs to Supermarket Cyble frst identified the listing which mentions "Bigbasket.com- -Customer Damabase" has

Cyble first identified the listing which mentions "Bigbasket.com - Customer Database" and claims it claims it was "dumped in was "dumped in late October 2020". Although BigBasket stated it does not store sensitive financial detai- cails like credit card numbers, the compromised personal informa- tion poses serious risks, including phishing and identity theft for millions of Indian customers.

The Bengaluru-based company confirmed the breach and is cur- rently investigating the depth of depth of the intrusion.

A police complaint has been filed with the Cyber Crime Cell in Bengaluru.

STOCK UPDATES

Figure 5.1: Print media report on the exfiltration and illegal commercial sale of BigBasket's customer database (November 2020).

Primary Reference Sources: Cyble Cyber Threat Intelligence Blog Research; Bengaluru Cyber Crime Police Investigation Documentation; Economic Times and The Hindu technology business reports; BigBasket corporate statement (November 2020).

Conclusion: Common Lessons Across Case Studies

A comparative synthesis across the five studied incidents highlights overarching vulnerabilities and actionable defensive requirements:

- **Preventable Known Vulnerabilities:** Most devastating intrusions exploited unpatched software flaws (Equifax Apache Struts), trusted developer updates (SolarWinds SUNBURST), or obsolete legacy mechanisms (Indian banks' magnetic stripes). Rigorous patch lifecycle management remains foundational.
- **Extended Attacker Dwell Times:** Adversaries consistently maintained undetected network presence for months before discovery. Defensive strategy must prioritize active interior traffic inspection and anomaly detection alongside perimeter barriers.
- **Supply Chain and Third-Party Risks:** External vendors and intermediate service partners represent high-risk attack surfaces (Hitachi ATM switch, SolarWinds build pipelines). Rigorous supplier verification and zero-trust boundaries are mandatory.
- **Physical and Kinetic Consequences:** Stuxnet proved that digital weapons can cause real physical kinetic destruction, requiring complete isolation and out-of-band monitoring for operational technology.
- **Transparent Crisis Disclosure:** Prompt communication with customers and regulatory authorities significantly reduces long-term commercial and legal harm following an unavoidable breach.

References

- BBC News, Reuters, Wired, The Hindu, Times of India, Economic Times, NDTV, Hindustan Times (Comprehensive news archives 2010–2021).
- US Federal Trade Commission (FTC), *Equifax Data Breach Settlement Statement* (2019); US House Oversight Committee Investigation Report (2018).
- Cybersecurity and Infrastructure Security Agency (CISA), *Emergency Directive 21-01: Mitigate SolarWinds Orion Code Compromise*; FireEye / Mandiant threat actor dossiers.
- Symantec Security Operations, *W32.Stuxnet Dossier Comprehensive Analysis* (February 2011).
- Reserve Bank of India (RBI) and National Payments Corporation of India (NPCI) official releases regarding the 2016 card network malware incidents.
- Cyble Research Labs, *BigBasket Data Leak Technical Bulletin* (November 2020).